

Abstract Algebra – Dummit & Foote

Notes by Gene

2026

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Part I

Group Theory

1 Introduction to Groups

1.1 Axioms

If $\star$ is a function on set G such that $\star : G \times G \rightarrow G$, then $\star$ is a **binary operation** on G .

- Associative if $\forall a, b \in G, a \star (b \star c) = (a \star b) \star c$

- Commutative if $\forall a, b \in G, a \star b = b \star a$

A **group** is an ordered pair $(G, \star)$ such that

1. $\star$ is associative
2. Existence of identity: $\exists e \in G$ such that $\forall a \in G, a \star e = e \star a = a$
3. Existence of inverse: $\forall a \in G, \exists a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$

*A group is **Abelian (commutative)** if $\forall a, b \in G, a \star b = b \star a$

E.g. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are groups under $+$ with identity $e = 0$ and inverse $a^{-1} = -a$. But $\mathbb{Z} - \{0\}$ is not a group under $\times$ because inverses are not guaranteed to exist.

Let $(G, \star)$ be a group, and $x \in G$. The **order** of x is the smallest positive integer n such that $x^n = 1$, and is denoted $|x|$. If no positive powers of x is 1, $|x| = \infty$.

- In the additive group $\mathbb{Z}/9\mathbb{Z}$, $\bar{6}$ has order 3, since $\bar{6} + \bar{6} + \bar{6} = \bar{18} = \bar{0}$.
- In the multiplicative group $\mathbb{R} - \{0\}$, -1 has order 2 since $(-1)^2 = 1$.

1.2 Dihedral Groups

For each $n \in \mathbb{Z}^+, n \geq 3$, let D_{2n} be the set of symmetries of a regular n -gon. E.g. $s \in D_{2n}$ might be a rotation of $2\pi/n$ around the n -gon.

$|D_{2n}| = 2n$, so we call D_{2n} the **dihedral group of order $2n$** .

Any symmetry can be written in terms of rotations (r) and reflections (s) alone, e.g. for $n = 12$,

$$(sr^9)(sr^6) = s(r^9s)r^6 = s(sr^{-9})r^6 = s^2r^{-3} = r^9$$

S is a set of generators of G if any element of G can be written as a finite product of the elements of S . This is notated:

$$G = \langle S \rangle$$

For the dihedral group, $S = \{s, r\}$. The equations in G that the generators satisfy are called **relations** in G . For D_{2n} , the relations are $r^n = 1$, $s^2 = 1$, and $rs = sr^{-1}$.

If G is generated by S , and there exists a collection of relations $R_1, R_2, \dots, R_m$, then these generators and relations are called the **presentation of G** :

$$G = \langle S \mid R_1, R_2, \dots, R_m \rangle$$

For the dihedral group, $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$.

1.3 Symmetric Groups

Let S_Ω be the bijection from Ω to itself (permutation). S_Ω is a group under the function composition $\circ$.

- Call $(S_\Omega, \circ)$ the **symmetric group** on set Ω .

- If $\Omega = \{1, 2, 3, \dots, n\}$, then we denote S_Ω as S_n , the **symmetric group of degree n** .
- $(S_n, \circ)$ has degree $n!$ ($n!$ elements).

We can represent S_n in a cycle decomposition, grouping disjoint cycles. E.g.

$$(a_1 \ a_2 \ \dots \ a_m)(a_{m_1+1} \ a_{m_1+2} \ \dots \ a_{m_2}) \cdots (a_{m_{k-1}+1} \ a_{m_{k-1}+2} \ \dots \ a_{m_k})$$

Meaning that a_1 goes to a_2 , a_{m_1+1} goes to a_{m_1+2} , etc. Values such that $\sigma(a_i) = a_i$ aren't written. For example, for $\sigma \in S_3$ with $\sigma(1) = 1$, $\sigma(2) = 3$, and $\sigma(3) = 2$, the cycle decomposition is

$$(2 \ 3)$$

1.4 Matrix Groups

First, we need to define a field. A field is a set F with 2 binary operators $+$ and $\cdot$ such that

1. $(F, +)$ and $(F - \{0\}, \cdot)$ are Abelian groups
2. The distribution law holds:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \quad \forall a, b, c \in F$$

Define the **general linear group of degree n** :

$$GL_n(F) = \{A \mid A \text{ is a } n \times n \text{ matrix with entries from } F, \text{ and } \det(A) \neq 0\}$$

- Since $\det(A) = 0$, $GL_n(F)$ is closed under matrix multiplication.
- Matrix products are associative.
- $\det(A) \neq 0$ iff. A has an inverse.

Thus, $GL_n(F)$ is a group under matrix multiplication.

1.5 The Quaternion Group

Simple enough:

$$\begin{aligned} Q_8 &= \{1, -1, i, -i, j, -j, k, -k\} \\ 1 \cdot a &= a \cdot 1 = a \\ (-1) \cdot a &= a \cdot (-1) = -a \\ i \cdot i &= j \cdot j = k \cdot k = -1 \\ i \cdot j &= k \quad , \quad j \cdot k = i \quad , \quad k \cdot i = j \\ j \cdot i &= -k \quad , \quad k \cdot j = -i \quad , \quad i \cdot k = -j \end{aligned}$$

$(Q_8, \cdot)$ is a non-Abelian group of order 8.

1.6 Homomorphisms and Isomorphisms

Let $(G, \star)$ and $(H, \diamond)$ be groups. A map $\phi : G \rightarrow H$ such that

$$\phi(x \star y) = \phi(x) \diamond \phi(y)$$

is a **homomorphism**.

- “A map that reflects the group structure of the domain and codomain.”

$\phi : G \rightarrow H$ is called an **isomorphism** if

1. ϕ is a homomorphism
2. ϕ is a bijection

Can also say: G and H are isomorphic; G and H are of the same isomorphism type; $G \cong H$.

I.e. G and H are the “same group” except the elements and operators are written differently (think abstract).

If $\phi : G \rightarrow H$ is an isomorphism, then

1. $|G| = |H|$
2. G is Abelian iff. H is Abelian
3. $\forall x \in G, |x| = |\phi(x)|$

1.7 Group Actions

A group action from group G on set A is a map from $G \times A$ to A satisfying

1. $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a \quad \forall g_1, g_2 \in G, \forall a \in A$
2. $1 \cdot a = a \quad \forall a \in A$

Let G act on A . For each $g \in G$, define σ_g to be the map

$$\begin{aligned} \sigma_g : A &\rightarrow A \\ \sigma_g(a) &= g \cdot a \end{aligned}$$

1. For each $g \in G$, σ_g is a permutation of A .
2. The map from G to S_A defined by $g \mapsto \sigma_g$ is a homomorphism.

I.e. every $g \in G$ acts as a permutation of A in a way consistent with the group operation in G .

Group actions treat $g \in G$ to be functions, sort of.

2 Subgroups

A subset H of a group G is a **subgroup** of G if

1. $H \neq \emptyset$
2. H is closed under products and inverses. I.e. $x, y \in H$ implies $x^{-1} \in H$ and $xy \in H$.

This is denoted $H \leq G$, or if $H \neq G$, $H < G$.

I.e. a subgroup is a subset of G that is also a group w.r.t the binary operation defined on G . A subgroup must thus contain the identity operator.

Every G has two subgroups: $H = G$ and $H = \{1\}$ (trivial subgroup).

A simpler subgroup criterion: a subset H of a group G is a subgroup of G iff.

1. $H \neq \emptyset$
2. $\forall x, y \in H, xy^{-1} \in H$

2.1 Centralizers and Normalizers, Stabilizers and Kernels

Let A be a subset of G .

The **centralizer of A in G** is the set of elements of G that commute with every element of A .

$$C_G(A) = \{g \in G \mid gag^{-1} = a, \forall a \in A\}$$

- For an Abelian group G , $C_G(A) = G, \forall A \subseteq G$.
- For the quaternion group, $C_{Q_8}(\{i\}) = \{\pm 1, \pm i\}$

The **center of G** is the set of elements of G commuting with every element of G .

$$Z(G) = C_G(G) = \{g \in G \mid gx = xg, \forall x \in G\}$$

The **normalizer of A in G** is the set of elements of G that commutes with the entire set A .

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}$$

This is a looser criterion than the centralizer, since order within A doesn't matter. Formally, if $g \in C_G(A)$, then $gag^{-1} = a$ for all $a \in A$, so $C_G(A) \leq N_G(A)$.

For an Abelian group G , which has the commutation property already, $C_G(A) = N_G(A) = Z(G) = G$, for any subset $A \subseteq G$.

If G is a group action on set S , and $s \in S$, the **stabilizer of s in G** is

$$G_s = \{g \in G \mid g \cdot s = s\}$$

- $1 \in G_s$

- If $y \in G_s$, then $y^{-1} \in G_s$.

$$\begin{aligned} s &= 1 \cdot s = (y^{-1}y) \cdot s \\ &= y^{-1} \cdot (y \cdot s) \\ &= y^{-1} \cdot s \end{aligned}$$

- If $x, y \in G_s$, then $xy \in G_s$.

$$\begin{aligned} (xy) \cdot s &= x \cdot (y \cdot s) \\ &= x \cdot s \\ &= s \end{aligned}$$

Thus G_s is a subgroup of the group action G , or $G_s \leq G$.

The **kernel of the action of G on set S** is

$$\{g \in G \mid g \cdot s = s, \forall s \in S\}$$

2.2 Cyclic Subgroups

A possible way to generate a subgroup H of G from an element $x \in G$ is to form the set $H = \{x^n \mid n \in \mathbb{Z}\}$. Guarantees closure under products and inverses.

A subgroup $H \subseteq G$ is **cyclic** if $\exists x \in H$ such that $H = \{x^n \mid n \in \mathbb{Z}\}$. I.e. H can be generated by 1 element. Notated $H = \langle x \rangle$. Cyclic groups are Abelian, by properties of the exponent.

- E.g. If $H = (\mathbb{Z}, +)$, then $H = \langle 1 \rangle$.
- E.g. If H is the subgroup of all rotations of an n -gon, then $H = \langle r \rangle$.

2.2.1 Theorems

1. If $H = \langle x \rangle$, then $|H| = |x|$.
2. Let $x \in G$. For $m, n \in \mathbb{Z}$, if $x^n = x^m = 1$, then $x^d = 1$, where $d = \gcd(m, n)$.
3. Any 2 cyclic groups of the same order are isomorphic.
 - If $n \in \mathbb{Z}^+$ and $\langle x \rangle, \langle y \rangle$ are cyclic groups of order n , the following map is an isomorphism.

$$\begin{aligned} \phi : \langle x \rangle &\rightarrow \langle y \rangle \\ x^k &\mapsto y^k \end{aligned}$$

- If $\langle x \rangle$ is an infinite cyclic group, the following map is a homomorphism. Thus for each $n \in \mathbb{Z}^+$, let Z_n be the cyclic group of order n .

$$\begin{aligned} \phi : \mathbb{Z} &\rightarrow \langle x \rangle \\ k &\mapsto x^k \end{aligned}$$

4. Let G be a map, $x \in G$, and $a \in \mathbb{Z} - \{0\}$.

- If $|x| = \infty$, then $|x^a| = \infty$
 - If $|x| = n < \infty$, then $|x^a| = n / \gcd(n, a)$
5. Let $H = \langle x \rangle$.
- If $|x| = \infty$, then $H = \langle x \rangle$ iff. $a = \pm 1$.
 - If $|x| = n < \infty$, then $H = \langle x^a \rangle$ iff. $\gcd(a, n) = 1$. The number of generators of H is $\phi(n)$.
 - E.g. $\bar{a}$ generates $\mathbb{Z}/n\mathbb{Z}$ iff. $(\bar{a}, n) = 1$. So $\bar{1}, \bar{5}, \bar{7}, \bar{11}$ are generators of $\mathbb{Z}/12\mathbb{Z}$, and these are all the $\phi(12) = 4$ generators.
6. Let $H = \langle x \rangle$ be a cyclic group. Then
- Every subgroup of H is cyclic.
 - If $|H| = \infty$, then for any non-negative and distinct $a, b \in \mathbb{Z}$, $\langle x^a \rangle = \langle x^b \rangle$.
 - If $|H| = n < \infty$, then for each $a \in \mathbb{Z}^+$ such that $a|n$, there's a unique subgroup of order a . This is the cyclic group $\langle x^d \rangle$, where $d = n/a$.

2.3 Subgroups Generated by Subsets

If $\mathcal{A}$ is any nonempty collection of subgroups of G , the intersection of all $\mathcal{A}$ is also a subgroup of G .

$$K = \bigcap_{H \in \mathcal{A}} H \text{ is a group if } H \leq G$$

If A is a subset of the group G , let the **subgroup of G generated by A** be

$$\langle A \rangle = \bigcap_{A \subseteq H, H \leq G} H$$

I.e. the intersection of all subgroups of G containing A . Equivalently,

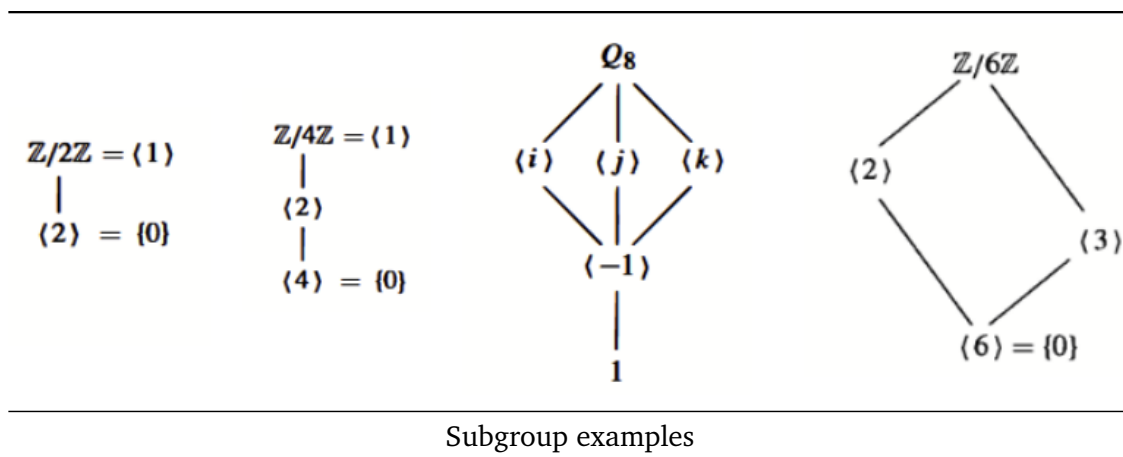
$$\langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \mid \forall i, a_i \in A, \alpha_i \in \mathbb{Z}, a_i \neq a_{i+1} \text{ and } n \in \mathbb{Z}^+\}$$

An Abelian group G simplifies this a bit:

$$\langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \mid \forall i \text{ and } \alpha_i \in \mathbb{Z}\}$$

- Without an Abelian structure, determining the generated subgroup becomes difficult. E.g. if $G = D_8$, let r, s be the generators. Let $a = s, b = rs$ so that $G = \langle a, b \rangle$. Thus G is generated by 2 elements of order 2, but has order 8.
- Or for $G = S_n = \langle (1\ 2), (1\ 2\ 3 \dots n) \rangle$. Generated by elements of order 2 and n , but has order $n!$

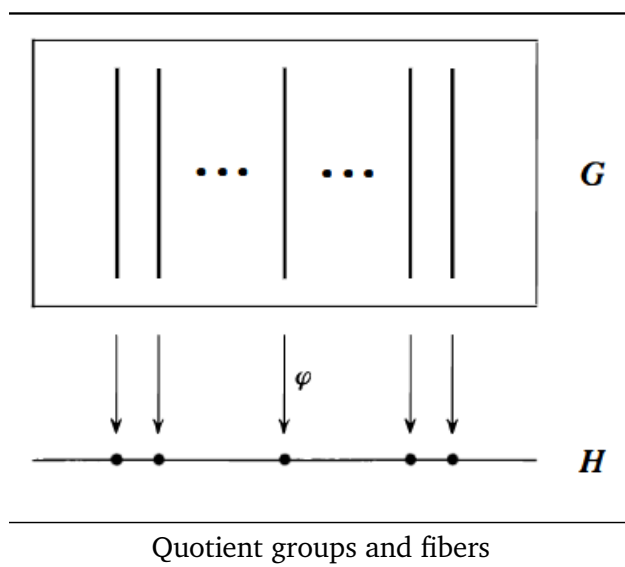
Lattice diagrams can help illuminate the structure of groups and subgroups. Build up from $\{0\}$ at the bottom, and draw subgroups of increasingly higher order upwards. This is helpful to see, for example, the smallest subgroup of G , $\langle H, K \rangle$, containing two smaller subgroups $H \leq G$ and $K \leq G$.



3 Quotient Groups and Homomorphisms

3.1 Definitions

Quotient groups are a way to obtain a smaller group from a group G .



$\phi : G \rightarrow H$ is a homomorphism from G to H , i.e. $\phi(xy) = \phi(x)\phi(y)$ for $x, y \in G$. Vertical lines above each point $h \in H$ represent the **fiber of ϕ over h** .

Denote the **kernel of ϕ** as the set

$$\ker \phi = \{g \in G \mid \phi(g) = 1_H\}$$

Let K be the kernel. The **quotient group/factor group** G/K (" $G \bmod K$ ") is a group whose elements are the fibers of ϕ .

For any subgroup $N \leq G$ and $g \in G$, define

$$\text{Left coset of } N \text{ in } G: gN = \{gn \mid n \in N\}$$

$$\text{Right coset of } N \text{ in } G: Ng = \{ng \mid n \in N\}$$

Call any element of a coset the **representative** of that coset. The fibers of a homomorphism $\phi : G \rightarrow H$ are the left or right cosets of the kernel (it may help to think of this graphically). The elements of G/K are the left cosets gK or the right cosets Kg with $g \in G$.

3.2 Theorems

Theorem 1: Let $\phi : G \rightarrow H$ be a homomorphism. Then

1. $\phi(1_G) = 1_H$
2. $\phi(g^{-1}) = \phi(g)^{-1}$
3. $\phi(g^n) = \phi(g)^n$
4. Kernel of ϕ is a subgroup of G : $\ker \phi \leq G$
5. Image of G under ϕ is a subgroup of H : $\text{im}(\phi) \leq H$

Proof: For (1), $\phi(1_G) = \phi(1_G 1_G) = 1_H 1_H$. For (2), $\phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(1_G) = 1_H$. Then (3) follows from induction.

For (4), note that $1_G \in \ker \phi$, so $\ker \phi$ is nonempty. Let $x, y \in \ker \phi$. Then $\ker \phi$ satisfies the subgroup criterion:

$$\phi(xy^{-1}) = \phi(x)\phi(y)^{-1} = 1_H$$

Similarly for (5), note that $\phi(1_G) = 1_H$, so the identity of H lies in $\text{im}(\phi)$. If $x, y \in \text{im}(\phi)$ with $\phi(a) = x$ and $\phi(b) = y$, then $\text{im}(\phi)$ satisfies the subgroup criterion:

$$\begin{aligned} xy^{-1} &= \phi(a)\phi(b)^{-1} = \phi(ab^{-1}) \\ xy^{-1} &\in \text{im}(\phi) \end{aligned}$$

Definition: Let $\phi : G \rightarrow H$ with kernel K . Then the **quotient group** G/K is the group whose elements are fibers of ϕ with group operations defined such that: if X is the fiber above a and Y is the fiber above b , then $X \cdot Y$ is the fiber above ab .

Thus $K \in G/K$. Other elements of G/K are “translations” of K .

E.g. $\mathbb{Z}/4\mathbb{Z}$ is the group with set $\{0, 1, 2, 3\}$ and group operation “addition mod 4.” The kernel $K = \{0\}$ is included in $\mathbb{Z}/4\mathbb{Z}$. ϕ could be the operation “modulo 4,” which has kernel $4\mathbb{Z}$.

Theorem 2: Let $\phi : G \rightarrow H$ be a homomorphism with kernel K . Let $X \in G/K$ be a fiber above a , i.e. $X = \phi^{-1}(a)$. Then

- 1) $\forall u \in X, X = \{uk \mid k \in K\}$
- 2) $\forall u \in X, X = \{ku \mid k \in K\}$

Proof: First show $uK \subseteq X$, then $X \subseteq uK$. Similarly for Ku .